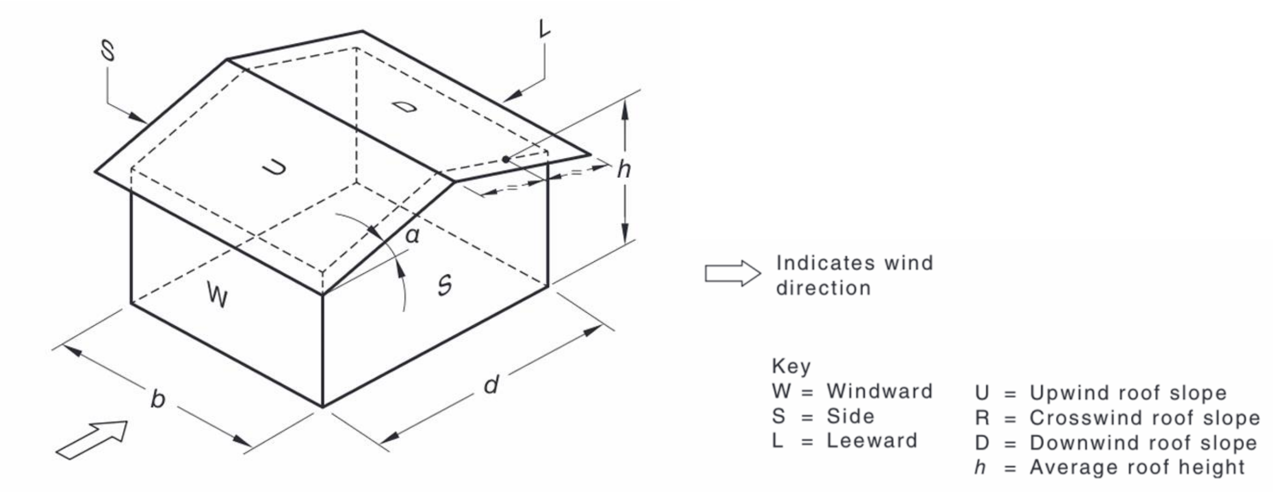
PART F

Case 1.

External pressure coefficients, $C_{p,e}$



$$L_R = \frac{L_p}{L_m} = 100 \Longrightarrow L_p = 100L_m$$

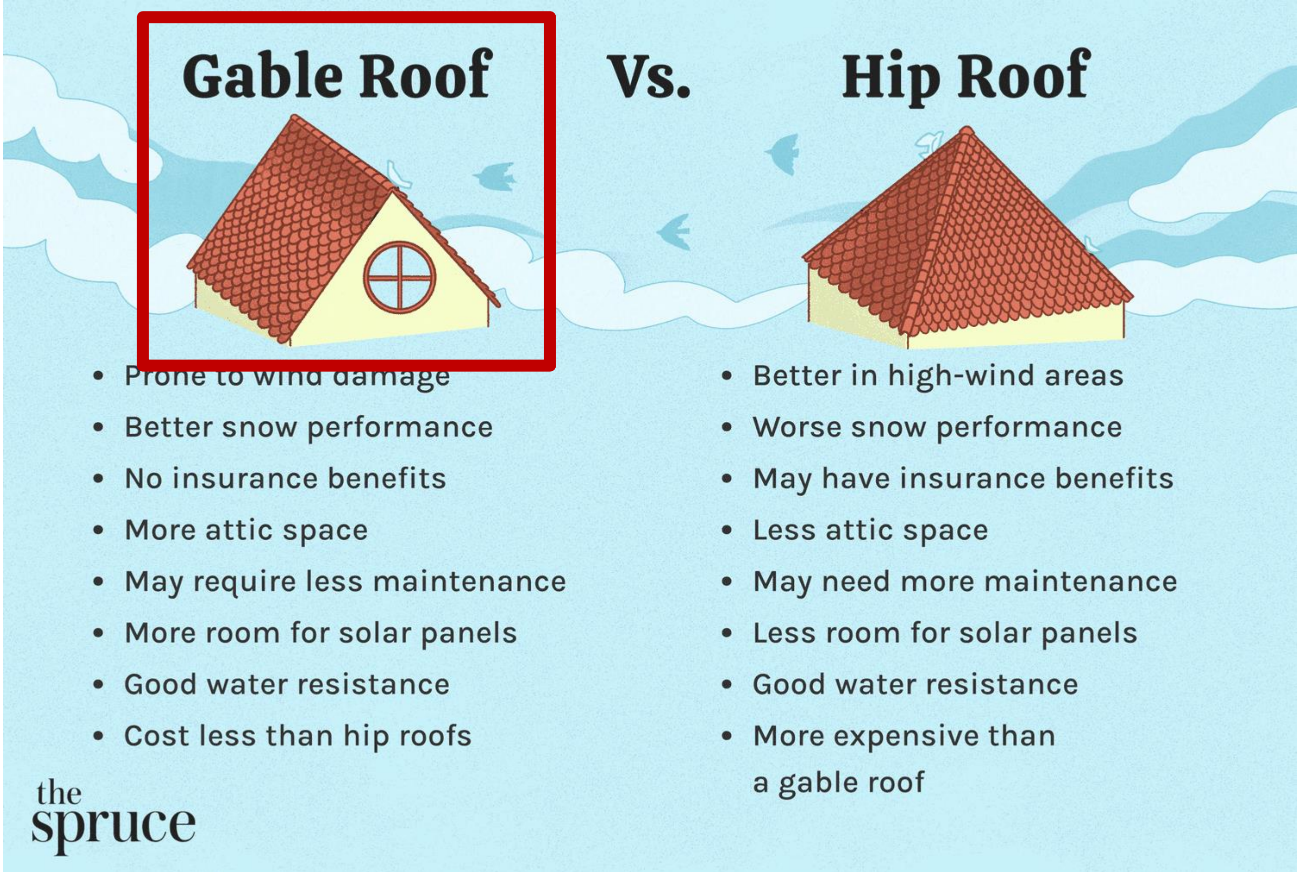
In the model:

$$b_m = 105\ mm \Longrightarrow b_p = 10.5\ m$$

$$d_m = 48\ mm \Longrightarrow d_p = 4.8\ m$$

$$h_m = 37 + \frac{53 - 37}{2} = 45\ mm \Longrightarrow h_p = 4.5\ m$$

Roof pitch: $\alpha = \tan^{-1}\left(\frac{1.6}{2.4}\right) = 33.69°$



Wall type	$C_{p,e}$	Reference
Windward wall (W)	0.7	Table 5.2(A)
Leeward wall (L)	-0.5	Table 5.2(B), $\theta = 0°$ Roof pitch $\alpha = 33.69° > 25°$ $\frac{d}{b} = \frac{4.8}{10.5} = 0.4571 > 0.3$
Sidewalls (S)	Horizontal distance from windward edge between 0 and 4.5m. $C_{p,e} = -0.65$ Horizontal distance from windward edge between 4.5m and 4.8m. $C_{p,e} = -0.5$	Table 5.2(C) When the horizontal distance from windward edge is between 0 and 4.5m, the distance is considered as 0 to 1h. When the distance is between 4.5m and 4.8m, the distance is between 1h and 2h.
Upwind slope (U)	$-0.2230, 0.2863$	Table 5.3(B) Roof pitch $\alpha = 33.69° > 10°$ $\frac{h_p}{d_p} = \frac{4.5}{4.8} = 0.9375$ Linear interpolation on α and h/d.
Downwind slope (D)	-0.6	Table 5.3(C) Roof pitch $\alpha = 33.69° > 25°$ $\frac{b_p}{d_p} = \frac{10.5}{4.8} = 2.1875 < 3$

Where two values of $C_{p,e}$ are listed, roofs shall be designed for both values. In these cases, roof surfaces may be subjected to either value due to turbulence. Alternative combinations of external and internal pressures (see also Clause 5.3) shall be considered, to obtain the most severe conditions for design.

In regard to Tables 5.3(A), 5.3(B) and 5.3(C), for intermediate values of h/d ratios, linear interpolation shall be used. Interpolation shall only be carried out on values of the same sign.

Table 5.3(B) — Roofs — External pressure coefficients ($C_{p,e}$) for rectangular enclosed buildings — Upwind slope (U) $\alpha \geq 10°$

Upwind slope (U)	Ratio h/d	External pressure coefficients ($C_{p,e}$)						
		Roof pitch (α) degrees						
		10	15	20	25	30	35	≥ 45
$\alpha \geq 10°$	≤ 0.25	−0.7, −0.3	−0.5, 0.0	−0.3, 0.2	−0.2, 0.3	−0.2, 0.4	0.0, 0.5	0, 0.8 sin α
	0.5	−0.9, −0.4	−0.7, −0.3	−0.4, 0.0	−0.3, 0.2	−0.2, 0.3	−0.2, 0.4	
	≥ 1.0	−1.3, −0.6	−1.0, −0.5	−0.7, −0.3	−0.5, 0.0	−0.3, 0.2	−0.2, 0.3	

A1 Table 5.3(C) — Roofs — External pressure coefficients ($C_{p,e}$) for rectangular enclosed buildings — Downwind slope (D), and (R) for hip roofs, for $\alpha \geq 10°$

Roof type and slope		Ratio h/d	External pressure coefficients ($C_{p,e}$)			
Crosswind slope for hip roofs (R)	Downwind slope (D)		Roof pitch (α), degrees			
			10	15	20	≥ 25
$\alpha \geq 10°$	$\alpha \geq 10$	≤ 0.25	−0.3	−0.5	−0.6	For $b/d \leq 3$; −0.6
		0.5	−0.5	−0.5	−0.6	For $3 < b/d < 8$; $-0.06\,(7 + b/d)$
		≥ 1.0	−0.7	−0.6	−0.6	For $b/d \geq 8$; −0.9 A1

Aerodynamic shape factor, C_{shp}

Only need to consider frictional drag forces and external pressures for enclosed buildings.

[Cl 5.2(a)]

External pressure: $C_{shp} = C_{p,e}K_aK_{c,e}K_\ell K_p$

Frictional drag forces: $C_{shp} = C_fK_aK_{c,e}$ (Not considered, see page 5)

External pressure

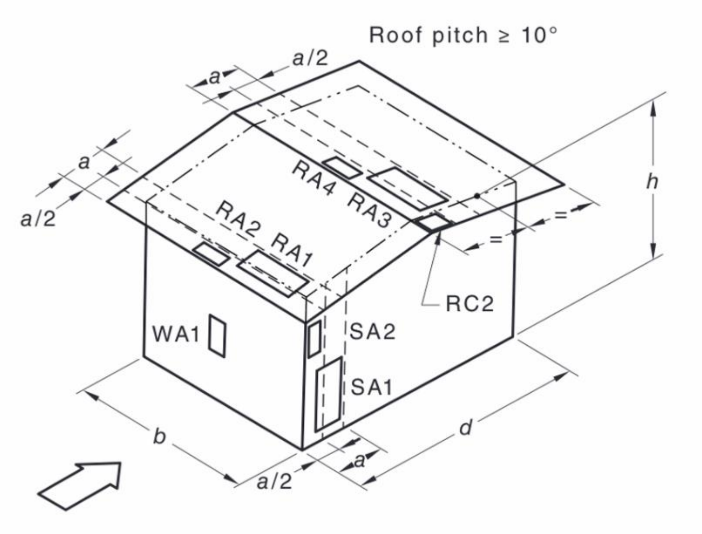
$C_{p,e}$ already found

K_a can be found by Table 5.4. with Tributary area $A < 1\ m^2$

	Roofs (U, D) and side walls (S)	Windward walls (W) $(h_p < 25\ m)$	Leeward walls (L) $(h < 25\ m)$
K_a	1.0	1.0	1.0

$K_{c,e} = 0.8$ (Table 5.5, Design case (a))

Local pressure factor K_ℓ for cladding [Cl 5.4.4]



The value of dimension a,

Walls: $a = \min(0.2b, 0.2d, h)$

Roofs:

1. $\frac{h_p}{b_p}$ or $\frac{h_p}{d_p} \geq 0.2$, $a = \min(0.2b, 0.2d)$
2. Both $\frac{h_p}{b_p}$ and $\frac{h_p}{d_p} < 0.2$, $a = 2h$.

Walls:

$$a = \min\left(0.2b_p, 0.2d_p, h_p\right) = \min(0.2 \times 10.5, 0.2 \times 4.8, 4.5) = 0.96$$

$$a^2 = 0.9216$$

$$0.25a^2 = 0.2304$$

Roofs:

$$\frac{h_p}{b_p} = \frac{4.5}{10.5} = 0.4286 > 0.2$$

$$a = \min\left(0.2b_p, 0.2d_p\right) = 0.96$$

The context only mentioned "tributary area of each fixing is less than 1 m2 in full scale". We consider the worst-case scenario here, which is $A \leq 0.25a^2$.

Design case	Reference case	Location (Proximity to edge)	K_ℓ
Windward wall (W)	WA1	Anywhere	1.5
Downwind corners of roofs with pitch ≥ 10°	RC2	< 0.96m from both roof edge and ridge	3.0
Upwind roof edges	RA1	< 0.96m	1.5
	RA2	< 0.48m	2.0
Downwind side of hips and ridges of roofs with pitch ≥ 10°	RA3	< 0.96m	1.5
	RA4	< 0.48m	2.0
Sidewall near windward wall edges	SA1	< 0.96m	1.5
	SA2	< 0.48m	2.0
All other areas	-	-	1.0

Permeable cladding reduction factor, K_p [Cl 5.4.5]

We have solid cladding, $K_p = 1.0$

Frictional drag forces [Cl 5.5]

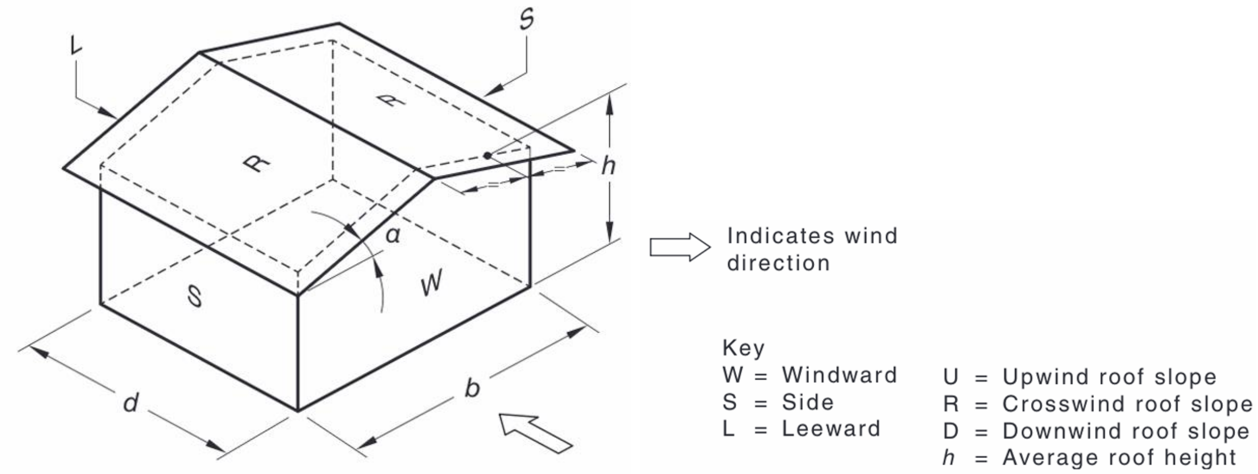
$$\frac{d_p}{h_p} = \frac{4.8}{4.5} = 1.067 < 4$$

$$\frac{d_p}{b_p} = \frac{4.8}{10.5} = 0.4571 < 4$$

No need to consider frictional drag forces hence neither $\frac{d}{h}$ nor $\frac{d}{b}$ is greater than 4.

Case 2.

External pressure coefficients, $C_{p,e}$



$$d_p = 10.5\ m$$

$$b_p = 4.8\ m$$

$$h_p = 4.5\ m$$

Roof pitch: $\alpha = \tan^{-1}\left(\frac{1.6}{2.4}\right) = 33.69°$

Wall type	$C_{p,e}$	Reference
Windward wall (W)	0.7	Table 5.2(A)
Leeward wall (L)	−0.29	Table 5.2(B), $\theta = 90°$ Roof pitch $\alpha = 33.69° > 25°$ $\frac{d}{b} = \frac{10.5}{4.8} = 2.1875$ Linear interpolation: $-0.3 + \frac{2.1875 - 2}{4 - 2}\left(-0.2 - (-0.3)\right) = -0.290625$

Sidewalls (S)	Horizontal distance from windward edge between 0 and 4.5m. $C_{p,e} = -0.65$	Table 5.2(C) When the horizontal distance from windward edge is between 0 and 4.5m, the distance is considered as 0 to 1h. When the distance is between 4.5m and 9m, the distance is between 1h and 2h. Between 9m and 10.5m is in the range of 2h to 3h.
	Horizontal distance from windward edge between 4.5m and 9m. $C_{p,e} = -0.5$	
	Horizontal distance from windward edge between 9m and 10.5m. $C_{p,e} = -0.3$	
Upwind slope (U)	$-0.1578, 0.4024$	Table 5.3(B) Roof pitch $\alpha = 33.69° > 10°$ $\frac{h_p}{d_p} = \frac{4.5}{10.5} = 0.4286$ Linear interpolation on α and h/d.
Downwind slope (D)	-0.6	Table 5.3(C) Roof pitch $\alpha = 33.69° > 25°$ $\frac{b_p}{d_p} = \frac{4.8}{10.5} = 0.4571 < 3$

Aerodynamic shape factor, C_{shp}

Only need to consider frictional drag forces and external pressures for enclosed buildings.

[Cl 5.2(a)]

External pressure: $C_{shp} = C_{p,e}K_aK_{c,e}K_{\ell}K_p$

Frictional drag forces: $C_{shp} = C_fK_aK_{c,e}$ (Not considered, see page 7)

External pressure

$C_{p,e}$ already found

K_a can be found by Table 5.4. with Tributary area $A < 1\ m^2$

	Roofs (U, D) and side walls (S)	Windward walls (W) $(h_p < 25\ m)$	Leeward walls (L) $(h < 25\ m)$
K_a	1.0	1.0	1.0

$K_{c,e} = 0.8$ (Table 5.5, Design case (a))

Local pressure factor K_ℓ for cladding [Cl 5.4.4]

Walls:

$$a = \min(0.2b_p, 0.2d_p, h_p) = \min(0.2 \times 10.5, 0.2 \times 4.8, 4.5) = 0.96$$

$$a^2 = 0.9216$$

$$0.25a^2 = 0.2304$$

Roofs:

$$\frac{h_p}{b_p} = \frac{4.5}{4.8} = 0.9375 > 0.2$$

$$a = \min(0.2b_p, 0.2d_p) = 0.96$$

The context only mentioned "tributary area of each fixing is less than 1 m2 in full scale". We consider the worst-case scenario here, which is $A \leq 0.25a^2$.

Design case	Reference case	Location (Proximity to edge)	K_ℓ
Windward wall (W)	WA1	Anywhere	1.5
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Upwind roof edges	RA1	< 0.96m	1.5
	RA2	< 0.48m	2.0
Downwind side of hips and ridges of roofs with pitch ≥ 10°	RA3	< 0.96m	1.5
	RA4	< 0.48m	2.0
Sidewall near windward wall edges	SA1	< 0.96m	1.5
	SA2	< 0.48m	2.0
All other areas	-	-	1.0

The building aspect ratio r is calculated as follows:

$$r = \frac{h_p}{\min(b_p, d_p)} = \frac{4.5}{\min(4.8, 10.5)} = \frac{4.5}{4.8} = 0.9375 < 1$$

Permeable cladding reduction factor, K_p [Cl 5.4.5]

We have solid cladding, $K_p = 1.0$

Frictional drag forces [Cl 5.5]

$$\frac{d_p}{h_p} = \frac{10.5}{4.5} = 2.3333 < 4, \qquad \frac{d_p}{b_p} = \frac{10.5}{4.8} = 2.1875 < 4$$

No need to consider frictional drag forces hence neither $\frac{d}{h}$ nor $\frac{d}{b}$ is greater than 4.